

Part A - First order Prediction

$$x[n] = \{10, 12, 15, 17, 18, 20, 21, 23\}$$

$$\hat{x}[n] = x[n-1]$$

A1. $\hat{x}[n] = \{-, 10, 12, 15, 17, 18, 20, 21\}$

* since $\hat{x}[n] = x[n-1]$ requires a previous sample, the predictor is undefined at $n=0$ (there's no $x=-1$). therefore, $\hat{x}[0]$ and $e[0]$ will be left blank

A2. $e[n] = \{-, 2, 3, 2, 1, 2, 1, 2\}$

$$\boxed{\text{rang} = \text{max} - \text{min}}$$

A3. $\text{Range}_{x[n]} = 23 - 10 = \underline{13}$ From 10 to 23

$\text{Range}_{e[n]} = 3 - 1 = \underline{2}$ From 1 to 3

A4. To quantize a signal properly, the number of levels used must be enough to cover its full dynamic range. Since $e[n]$ has a small range (2) compared to $x[n]$ (range 13), it needs far fewer quantization levels to achieve the same step size, and therefore fewer bits per sample.

Part B - Effect of the Predictor Coefficient

$$\hat{x}[n] = a x[n-1]$$

$$* x[n] = \{10, 12, 15, 17, 18, 20, 21, 23\}$$

For $a = 0$

starting
at $n=1$

$$\hat{x}[n] = \{0, 0, 0, 0, 0, 0, 0\}$$

$$e[n] = \{12, 15, 17, 18, 20, 21, 23\}$$

$$e^2[n] = \{144, 225, 289, 324, 400, 441, 529\}$$

$$\sum e^2 = 2352$$

$$MSE = \frac{1}{N-1} \sum e^2$$

$$= \frac{2352}{7} = \boxed{336}$$

For $a = 0.5$

$$\hat{x}[n] = \{5, 6, 7.5, 8.5, 9, 10, 10.5\}$$

$$e[n] = \{7, 9, 9.5, 9.5, 11, 11, 12.5\}$$

$$e^2[n] = \{49, 81, 90.25, 90.25, 121, 121, 156.25\}$$

$$\sum e^2 = 708.75$$

$$MSE = \frac{708.75}{7} = \boxed{101.25}$$

For $\alpha = 1$

$$\hat{X}[n] = \{10, 12, 15, 17, 18, 20, 21\}$$

$$e[n] = \{2, 3, 2, 1, 2, 1, 2\}$$

$$e^2[n] = \{4, 9, 4, 1, 4, 1, 4\}$$

$$\sum e^2[n] = 27$$

$$MSE = \frac{27}{7} = \boxed{3.86}$$

For $\alpha = 1.2$

$$\hat{X}[n] = \{12, 14.4, 18, 20.4, 21.6, 24, 25.2\}$$

$$e[n] = \{0, 0.6, -1, -2.4, -1.6, -3, -2.2\}$$

$$e^2[n] = \{0, 0.36, 1, 5.76, 2.56, 9, 4.84\}$$

$$\sum e^2 = 23.52$$

$$MSE = \frac{23.52}{7} = \boxed{3.36}$$

4. the smallest MSE is when $a = 1.2$
 $MSE = \underline{3.36}$

5. When $a = 0$, there is no prediction at all
 $e[n] = x[n]$

Prediction error carries exactly same information and dynamic range as the original signal, so we are quantizing $x[n]$ directly with no compression.

6. When the coefficient is excessively large, the predictor exceeds the true value of the signal, so the error no longer stays small and around zero. It grows in magnitude and can reach large negative values. This means the quantizer ends up handling wider dynamic range than the original signal itself, which is the opposite of what DPCM is meant to achieve.

7. No, prediction only reduces error's amplitude when a suitable coefficient is used. e.g:

at $a = 0 \rightarrow e[n] = x[n]$ zero reduction in amp.

at $a = 0.5 \rightarrow$ range of $e[n]$ (7 to 12.5), larger than $a = 1$ error range

~~So a reduction~~

So a good reduction in amp. is only achieved by

finding the optimum coefficient.

Part C - Finding the Optimum Predictor

$$a_{\text{opt}} = \frac{\sum x[n]x[n-1]}{\sum x^2[n-1]}$$

To find optimum a we need to calculate minimum

$$\text{MSE} = \frac{\sum e^2}{N-1}$$

since $(N-1)$ is const. $\rightarrow$ calculate min. $\sum e^2$

$$F(a) = \sum e^2$$

$$= \sum (x[n] - \hat{x}[n])^2$$

$$= \sum (x[n] - ax[n-1])^2$$

$$= \sum (x^2[n] - 2ax[n]x[n-1] + a^2x^2[n-1])$$

to find min. val. we diff. with respect to a and set the result to zero:

$$0 = \sum (0 - 2x[n]x[n-1] + 2ax^2[n-1])$$

$$0 = -\sum x[n]x[n-1] + \sum ax^2[n-1]$$

$$a_{\text{opt}} = \frac{\sum x[n]x[n-1]}{\sum x^2[n-1]}$$

a -opt for sequence

$$X_{[n]} = \{10, 12, 15, 17, 18, 20, 21, 23\}$$

$$X_{[n-1]} = \{-, 10, 12, 15, 17, 18, 20, 21\}$$

$$a\text{-opt} = \frac{\sum x_{[n]} x_{[n-1]}}{\sum x_{[n-1]}^2} \quad n=1 \rightarrow N-1$$

$$= \frac{12(10) + 15(12) + \dots}{(10)^2 + (12)^2 + \dots}$$

$$a\text{-opt} = \frac{2124}{1923} = 1.105$$

$$\Rightarrow \hat{X}_{[n]} = \{11.05, 13.26, 16.575, 18.785, 19.89, 22.1, 23.205, 25.415\}$$

$$e_{[n]} = \{0.95, 1.74, 0.425, -0.785, 0.11, -1.1, -0.205\}$$

$$e_{[n]}^2 = \{0.902, 3.027, 0.181, 0.616, 0.012, 1.21, 0.042\}$$

$$\sum e^2 = 5.99$$

$$MSE = \frac{5.99}{7} = 0.856$$

a	0	0.5	1	1.2	1.105 (a -opt)
MSE	336.	101.25	3.857	3.36	0.856

As we can see a -opt beats every value tested in Part B.